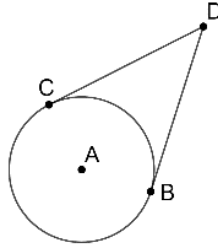


Geometry
Unit 12
Lesson 6 Practice

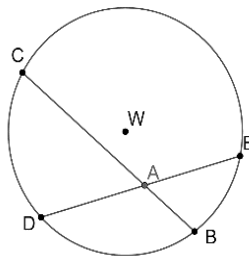
Name _____
Date _____
Period _____

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 10x + 8$ and $DB = 15x - 12$.



1. What is the value of x ?
2. What is the length of $\overline{CD}$?
3. What is the length of $\overline{DB}$?

For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A .
 $CA = 4x - 5$, $AB = 4$, $DA = 6$, and $AE = 2x + 2$.



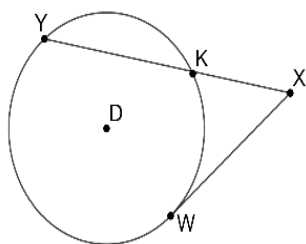
4. Write an equation you can use to solve for x .

5. Solve the equation you wrote in question 4 for x .

6. What is the length of $\overline{CB}$?

7. What is the length of $\overline{DE}$?

For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle and intersects tangent $\overline{XW}$ at point X . $XW = 10$, $KX = 4$, and $YX = 5x + 10$.



8. Write an equation that can be used to find the value of x .

9. Solve the equation you wrote in question 8 to find the value of x .

10. What is the length of $\overline{YK}$?